



Interdisciplinary Deep Learning On Graphs

Course Project Review

Node-to-Self

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Interdisciplinary Deep Learning on Graphs

PromptGFM-Bio: Prompt-Conditioned Graph Foundation Model for Rare-Disease Gene Mapping



- Rare diseases affect over 300 million people worldwide, yet most have no effective treatment — the core bottleneck being **gene discovery**: identifying which genes cause a disease. For most rare diseases, only 1–5 causal genes are known, making this a severe long-tail learning problem where standard supervised methods fail.
- **PromptGFM-Bio** addresses this by framing gene discovery as a link prediction problem on a large heterogeneous biomedical knowledge graph (19,576 genes, 16,841 diseases, 1.85M protein-protein interaction edges). The model combines a **GraphSAGE GNN backbone** with a **frozen PubMedBERT prompt encoder**, connected via **FiLM conditioning** — meaning a disease's natural language description dynamically modulates the GNN's message passing at every layer, not just at the final prediction step. This causes the same biological graph to produce different gene representations depending on the disease being queried.
- Given any disease description as input, the model ranks all candidate genes by their predicted association strength. The current best model achieves **AUROC 0.9626** and places the true causal gene in the top 50 (out of 19,576) for **55.7% of rare disease queries** — a 392× enrichment over random chance. The model also supports **zero-shot generalization** to diseases with no known gene associations seen during training.

What is unique about this project ?

- **FiLM-Conditioned Message Passing:** Disease text dynamically modulates GNN feature transformations at every layer via learned scale (γ) and shift (β) parameters — the language signal enters *inside* graph reasoning, not as static input features or at the prediction head. No prior work does this for gene prioritization.
- **Free-Text Disease Queries:** Unlike SHEPHERD (which requires structured HPO codes) or traditional KGE methods (which need disease IDs), PromptGFM-Bio accepts arbitrary natural-language disease descriptions via frozen PubMedBERT, enabling zero-shot generalization to diseases never seen during training.
- **Disease-Adaptive Graph Traversal:** The same PPI network produces *different* gene representations for different disease prompts — the model learns to "look at the graph differently" depending on which disease is being queried. Prior GNN methods (Mantis-ML 2.0, SHEPHERD, KDGene) generate disease-agnostic gene embeddings.

What is unique about this project ?

- **Full-Vocabulary Gene Ranking:** Evaluation ranks all ~19,576 candidate genes per disease query, not a small candidate set (SHEPHERD: 13–244 genes) or a few negatives. This is rare in the literature and provides an honest assessment of clinical utility.
- **Heterogeneous Biomedical Knowledge Graph:** Integrates STRING PPI (1.85M edges), DisGeNET, HPO phenotype bridges, and Orphanet into a single heterogeneous graph with 19,576 genes, 16,841 diseases, and 11,794 phenotypes.
- **117 Zero-Shot Rare Diseases:** Identifies diseases with zero training edges for evaluating true generalization — testing whether disease text alone can guide gene discovery without any prior labeled associations.
- **Combined Ranking-Aware Loss:** Jointly optimizes BCE, pairwise margin ranking, and ListNet losses for direct optimization of gene ranking quality, not just classification accuracy.

Data Sources

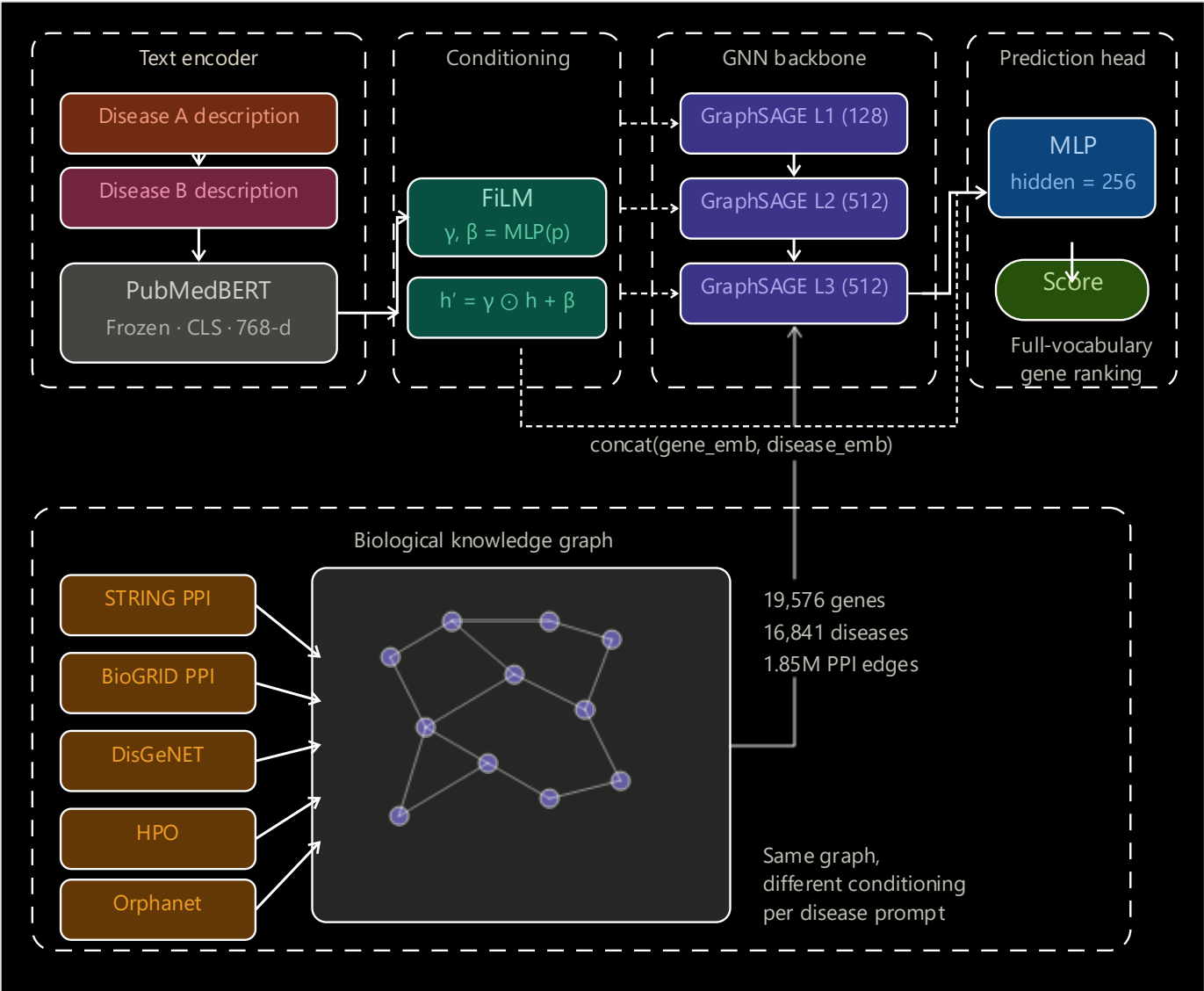
- **STRING v12** (PPI network) → Human protein–protein interactions (score ≥ 9606)
- **BioGRID 4.4.224** (PPI network) → Experimentally validated interactions
- **DisGeNET** (Gene–disease) → Curated gene–disease associations
- **HPO (Human Phenotype Ontology)** (Phenotype) → Gene–phenotype mappings, disease ontology
- **Orphanet** (Rare disease) → Rare disease gene associations

Graph Statistics

- **Gene nodes:** 19,576
- **Disease nodes:** 16,841
- **Phenotype nodes:** 11,794
- **Gene $\leftrightarrow$ Gene edges** (STRING PPI ≥ 700): 1,854,012
- **Gene $\rightarrow$ Disease edges:** 9,741,610
- **Disease $\rightarrow$ Gene edges** (reverse): 9,741,610

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Architecture



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Evaluation Metrics So Far



Variant	use_gnn	use_conditioning	AUROC	Hit@50	MRR	Description
MLP only (Baseline)	X	X	~0.955 (To be re-evaluated)	TBD	TBD	Raw node embeddings → MLP predictor. No message passing, no prompt conditioning.
Prompt only	X	✓	~0.956	TBD	TBD	FiLM conditioning active, but no PPI edges. Prompt signal enters at prediction only.
GNN only	✓	X	TBD	TBD	TBD	PPI message passing active, no FiLM ($\gamma=1$, $\beta=0$). GraphSAGE without language guidance.
Full PromptGFM-Bio	✓	✓	0.9626	0.5573	0.159	Full model with prompt-conditioned message passing over PPI graph.

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Project Progress So far



- **Constructed large-scale biomedical knowledge graph**
 - 19,576 genes, 16,841 diseases, 1.85M+ PPI edges (STRING + BioGRID)
- **Integrated multi-source biological datasets**
 - DisGeNET, HPO, Orphanet → unified gene–disease supervision (~1.17M edges)
- **Implemented prompt-conditioned GNN architecture (PromptGFM-Bio)**
 - GraphSAGE backbone + FiLM conditioning
 - Frozen PubMedBERT for disease text embeddings
- **Enabled disease-specific message passing**
 - Same graph produces different gene embeddings per disease query
- **Designed full-vocabulary evaluation protocol**
 - Rank all ~19.5k genes per disease (realistic, publication-grade setup)
- **Achieved strong performance on rare disease gene prioritization**
 - AUROC: **0.9626**, Hit@50: **55.7%**, MRR: **0.159**
- **Built ablation framework (4 variants × 3 seeds)**
 - Isolates contribution of GNN vs prompt conditioning
- **Identified 117 zero-shot rare diseases**
 - Enables evaluation of generalization to unseen diseases
- **Optimized training pipeline**
 - Mixed precision (AMP), stable training, ~2.6h per full run on RTX 4090



THANK YOU

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